

# CIFAR-10 exploration: complete record

Aubin Charley, Alexandre Corrad, Idriss El khamlichi, Maxime Nicaise

Companion archive of the working paper

This archive keeps the exhaustive view of the exploration summarized in the appendices of the working paper: every trained configuration, including negative results and runs that failed numerically, the asset sets that define pairing, and the figures covering whole campaigns. Accuracy is the final record of each run, without best-epoch selection; means and sample standard deviations are over valid seeds, and a single-seed configuration has no SD. Runs are paired only within matching asset sets and run conditions. The test set was used repeatedly, so these numbers are exploratory. Numbers come from the audited experiment index on branch **benchmark-organized** at commit 8d353f2 (19 experiments, 165 configurations, 377 runs); per-output provenance is in `paper/provenance/generated_assets.json`.

## 1 Pairing

Table 1: Pinned asset sets, identified by SHA-256 prefixes of their content. Pairing between runs is decided by these digests, not by batch names.

| Asset set             | Subset   | Probe    | Order, seeds 0 / 1 / 2         | Initial state, seeds 0 / 1 / 2 |
|-----------------------|----------|----------|--------------------------------|--------------------------------|
| r20bn-campaign-assets | 33236cc6 | a33645e0 | 5df32d0d / b3c1460f / 995dcd74 | 6e652c49 / f6fb2208 / fb19c69f |
| r20bn-ablation-assets | 33236cc6 | a33645e0 | 5df32d0d / b3c1460f / 995dcd74 | d8abf199 / 5c27c5f8 / f6b02312 |

Table 2: Plain ResNet-20 arms that verified identical pinned assets (initial weights, BatchNorm buffers, training probe, per-epoch order) in four batches. Final test accuracy (%). Evaluations before training are identical; the trajectories differ from the first later record. The cause is not established, so no correction is applied between batches.

| Experiment               | Seed 0 | Seed 1 | Seed 2 |
|--------------------------|--------|--------|--------|
| fulldata_gaussian        | 75.10  | 75.26  | 75.85  |
| campaign_grid21          | 75.13  | 75.62  | 76.10  |
| resbench_resolution_only | 75.31  | 75.34  | 75.95  |
| unified_selected         | 74.58  | 75.66  | 76.05  |
| Max – min                | 0.73   | 0.40   | 0.25   |

## 2 Resolution-only batch

This batch contains 29 targeted configurations with three seeds each, not a full factorial design: seven operators at the input and after block 1, max-pooling at five locations, four schedules for three operators after block 1, a plain arm and two fixed-resolution controls. Resolution-only means without an additional Gaussian continuation; smoothing built into an operator is part of the operator.

Table 3: Resolution-only batch (no additional Gaussian continuation): reduction operator at two locations, schedule  $16 \rightarrow 24 \rightarrow 32$ . Final test accuracy after 30 epochs, mean  $\pm$  sample SD over three seeds. Plain arm of the batch:  $75.53 \pm 0.36\%$ . The perceptual operator after block 1 produced a non-finite loss in all three seeds.

| Reduction operator            | After block 1 (%)    | At the input (%) |
|-------------------------------|----------------------|------------------|
| Adaptive max-pool             | $80.60 \pm 0.20$     | $79.23 \pm 0.48$ |
| MaxBlur (max, binomial blur)  | $80.43 \pm 0.75$     | $79.17 \pm 0.43$ |
| SoftPool                      | $80.34 \pm 0.48$     | $79.28 \pm 0.46$ |
| Bilinear (antialiased)        | $80.23 \pm 0.38$     | $79.56 \pm 0.68$ |
| $L^2$ reconstruction          | $80.11 \pm 0.52$     | $79.64 \pm 0.33$ |
| $\dot{H}^{-1}$ reconstruction | $79.85 \pm 0.76$     | $79.76 \pm 0.65$ |
| Perceptual (SSIM-style)       | Diverged (0/3 valid) | $79.98 \pm 0.40$ |

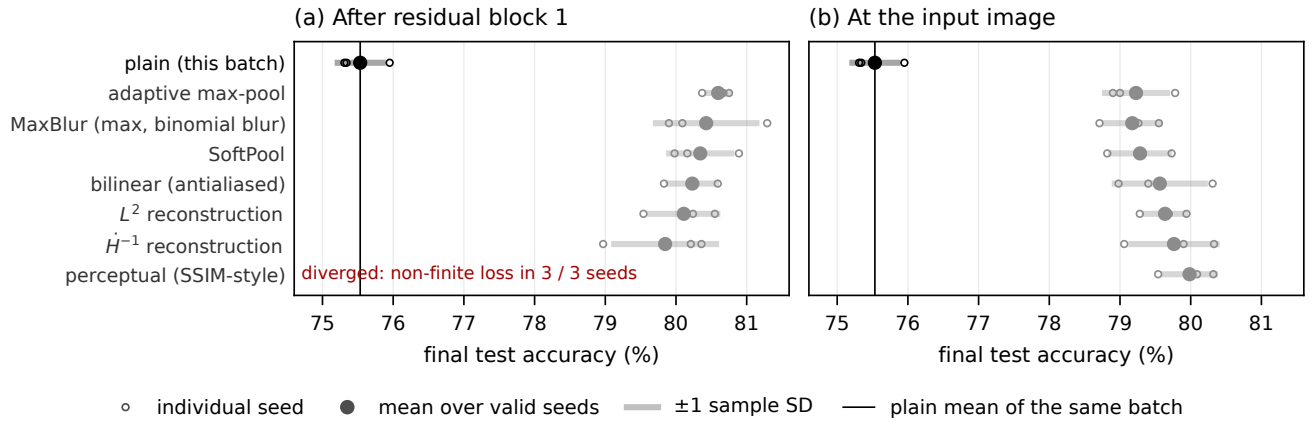


Figure 1: Operators after block 1 and at the input image, schedule  $16 \rightarrow 24 \rightarrow 32$ .

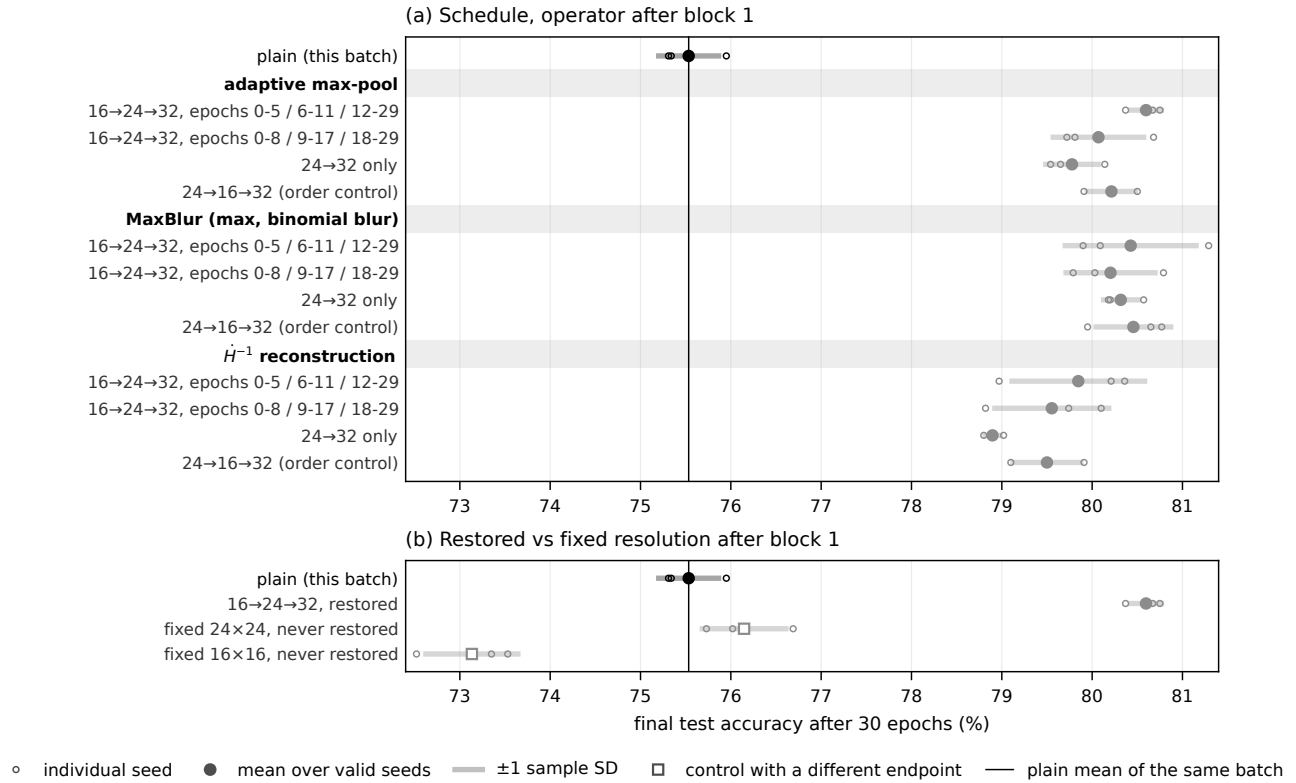


Figure 2: (a) Schedules after block 1 for three operators. (b) Progressive against fixed reductions; squares mark controls evaluated with their reduced network.

### 3 Other campaigns

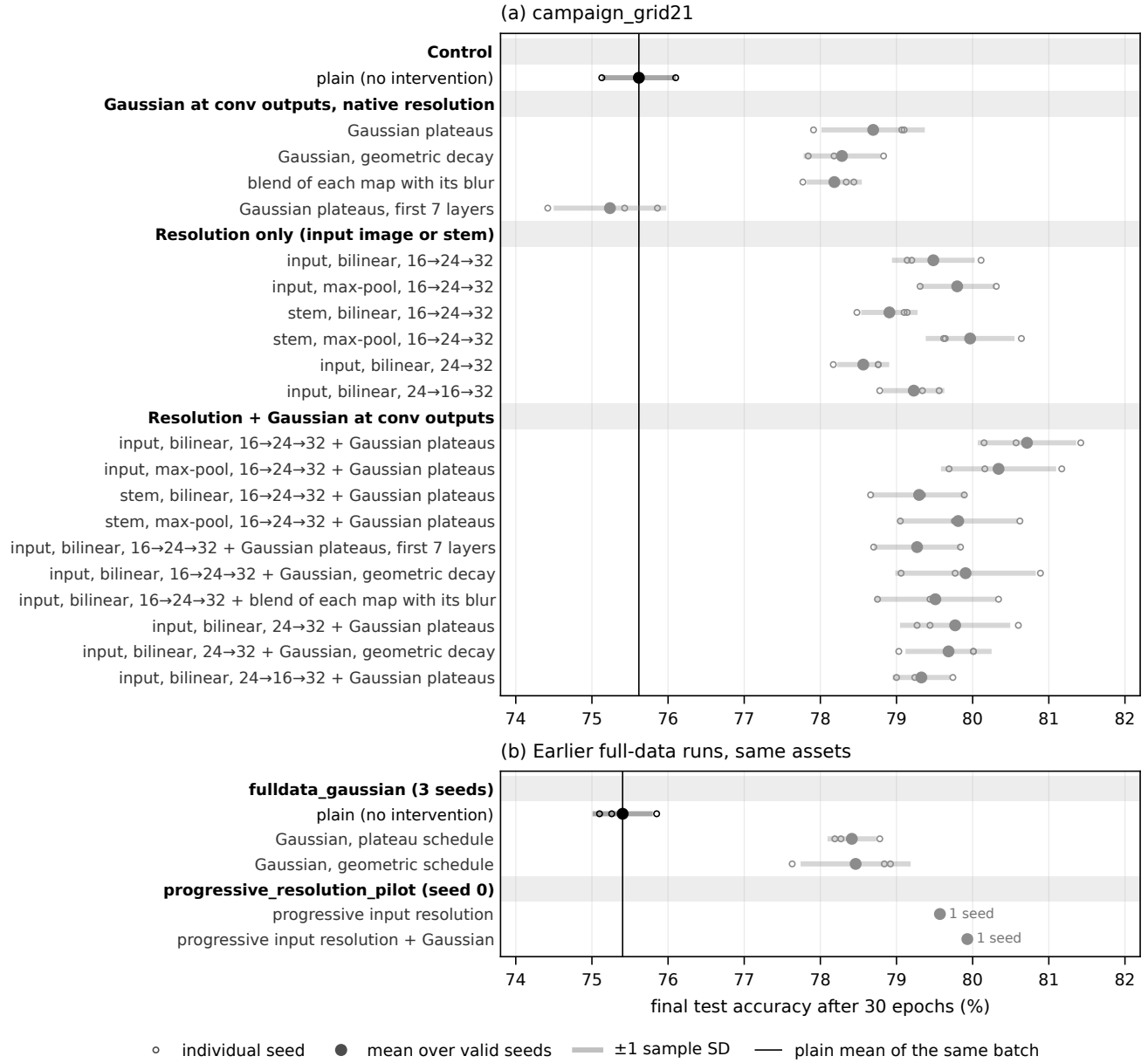


Figure 3: (a) The 21-configuration grid (input or stem reduction, Gaussian at convolution outputs). (b) Earlier full-data runs on the same pinned assets; the progressive-resolution pilot has one seed.

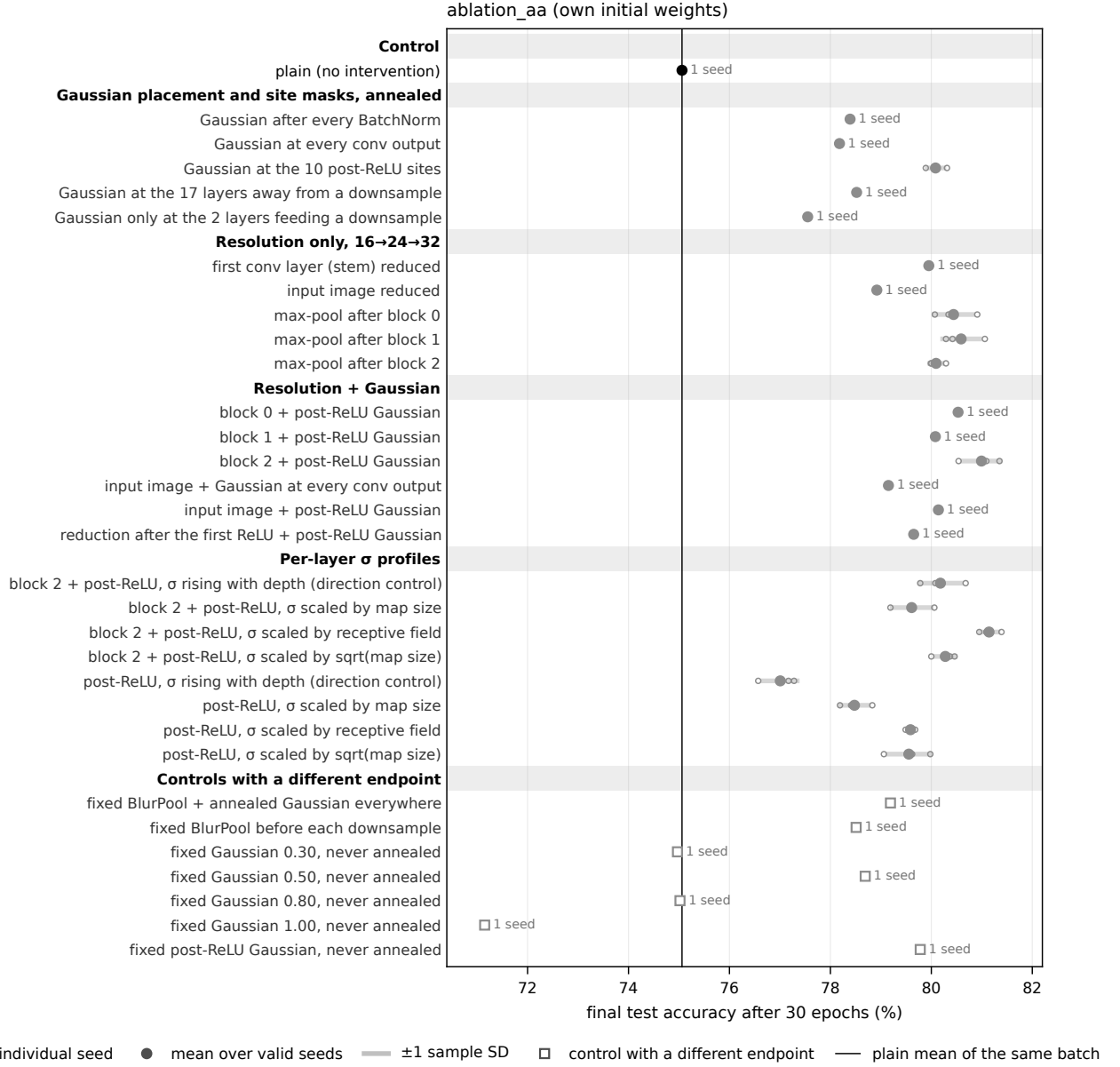


Figure 4: Ablation batch with its own initial weights and the campaign data order. Many arms have one seed. Squares mark controls that keep a fixed filter or BlurPool at inference.

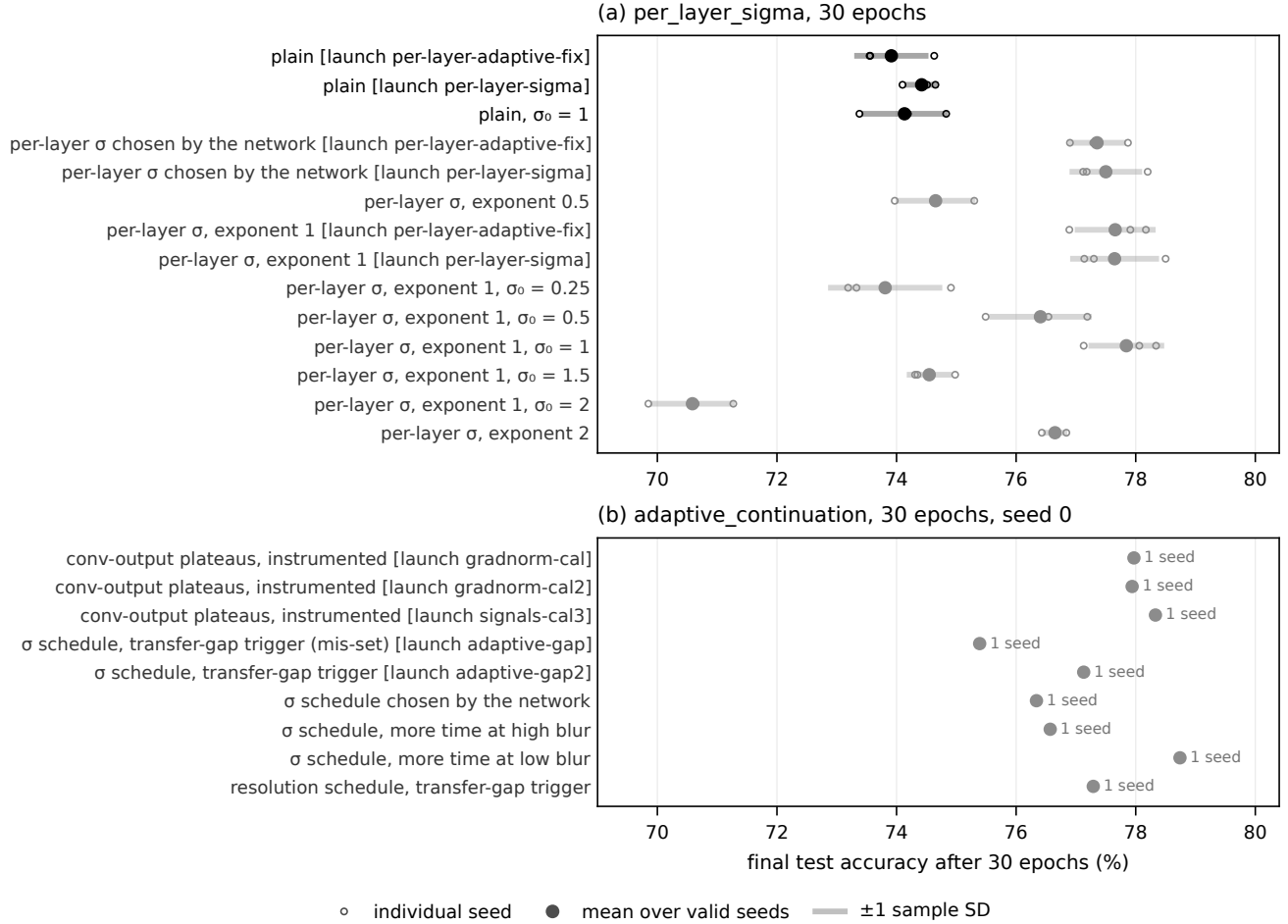


Figure 5: (a) Per-layer  $\sigma$  profiles on self-paired assets, two launches kept separate. (b) Data-triggered and reallocated schedules at seed 0. CPU pilots, smoke runs and 120-epoch runs are in the catalogue below.

## 4 Complete configuration catalogue

Table 4: Complete record of trained configurations, grouped by experiment. One row per configuration and run conditions; reruns under a separate launch are separate rows, while byte-identical copies of a summary are collapsed. Accuracy: final record of each run, mean over numerically valid seeds, no best-epoch selection. SD: sample SD in points; *n/a* with one valid seed. Seeds: valid / attempted. Endpoint: *target* is the native, filter-free path; *current (control)* marks arms whose intervention remains at inference. Wall: mean seconds per run, timing scope in the experiment header. Rows are alphabetical, not ranked; runs are paired only within matching asset sets and run conditions.

| Configuration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Config id            | Acc. (%) | SD   | Seeds | Endpoint | Wall (s) |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|----------|------|-------|----------|----------|
| <b>Promising methods in one batch, evaluated every epoch (unified_selected; owner Max; training benchmark)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |          |      |       |          |          |
| Conditions: reference recipe; evaluation after every epoch; per-epoch mean training loss recorded. Evaluation split: test. Records: benchmark-organized@8d353f2; results committed in 3829fa3. Shipped code checked against commit 3829fa3. Knowledge-base record: none. Assets: <b>r20bn-campaign-assets</b> : shares initial weights, BatchNorm buffers, probe and per-epoch order, seed by seed, with every experiment on this set. Runs attempted per manifests: 48; with summary: 48; numerically valid: 48; diverged: 0; failure markers: 0; duplicate summaries collapsed: 0. Timing: wall_seconds from model construction to summary write, including evaluations and checkpoint writes; eval_seconds counts snapshot evaluation only; train_seconds = wall - eval (so it includes data indexing and checkpoint I/O). One T4, two cells in parallel on a two-GPU kernel.                        |                      |          |      |       |          |          |
| Baseline: no blur, full 32x32 throughout                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | plain                | 75.43    | 0.76 | 3/3   | target   | 505      |
| Blur after every ReLU, annealed to zero                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | blur_relu            | 79.91    | 0.27 | 3/3   | target   | 624      |
| Blur every conv layer, annealed to zero                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | blur_conv            | 78.37    | 0.77 | 3/3   | target   | 737      |
| Shrink block 0 with max-pool, 16 to 24 to 32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | shrink_b0            | 80.24    | 0.48 | 3/3   | target   | 483      |
| Shrink block 0, plus blur after every ReLU                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | shrink_b0_relu       | 80.80    | 0.41 | 3/3   | target   | 623      |
| Shrink block 0, plus blur scaled by receptive field after every ReLU                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | shrink_b0_relu_rf    | 81.11    | 0.37 | 3/3   | target   | 620      |
| Shrink block 1 with max-pool, 16 to 24 to 32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | shrink_b1            | 80.37    | 0.61 | 3/3   | target   | 499      |
| Shrink block 1, plus blur after every ReLU                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | shrink_b1_relu       | 81.00    | 0.43 | 3/3   | target   | 631      |
| Shrink block 1, plus blur every conv layer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | shrink_b1_conv       | 81.51    | 0.22 | 3/3   | target   | 733      |
| Shrink block 1, plus blur scaled by receptive field after every ReLU                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | shrink_b1_relu_rf    | 81.05    | 0.57 | 3/3   | target   | 624      |
| Shrink block 1, plus blur scaled by sqrt of map size after every ReLU                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | shrink_b1_relu_sqrt  | 80.10    | 0.20 | 3/3   | target   | 636      |
| Shrink block 2 with max-pool, 16 to 24 to 32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | shrink_b2            | 80.12    | 0.45 | 3/3   | target   | 499      |
| Shrink block 2, plus blur after every ReLU                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | shrink_b2_relu       | 80.95    | 0.25 | 3/3   | target   | 620      |
| Shrink block 2, plus blur scaled by receptive field after every ReLU                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | shrink_b2_relu_rf    | 80.94    | 0.17 | 3/3   | target   | 621      |
| Shrink the input image, 16 to 24 to 32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | shrink_input         | 79.28    | 0.53 | 3/3   | target   | 482      |
| Shrink the input image, plus blur after every ReLU                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | shrink_input_relu    | 80.92    | 0.38 | 3/3   | target   | 619      |
| <b>Resolution-only batch: 29 targeted configurations (operator at two locations, max-pool at five locations, schedules for three operators, controls), no additional Gaussian continuation (resbench_resolution_only; owner Max; training benchmark)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |          |      |       |          |          |
| Conditions: reference recipe; evaluation every 2 epochs; fixed16 / fixed24 controls report their reduced-resolution path. Evaluation split: test. Records: benchmark-organized@8d353f2; results committed in c8c6804. Shipped code checked against commit 61b58e7. Knowledge-base record: none. Assets: <b>r20bn-campaign-assets</b> : shares initial weights, BatchNorm buffers, probe and per-epoch order, seed by seed, with every experiment on this set. Runs attempted per manifests: 87; with summary: 87; numerically valid: 84; diverged: 3; failure markers: 0; duplicate summaries collapsed: 0. Timing: wall_seconds from model construction to summary write, including evaluations and checkpoint writes; eval_seconds counts snapshot evaluation only; train_seconds = wall - eval (so it includes data indexing and checkpoint I/O). One T4, two cells in parallel on a two-GPU kernel. |                      |          |      |       |          |          |
| Baseline: no blur, full 32x32 throughout (resolution batch)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | none__none__none     | 75.53    | 0.36 | 3/3   | target   | 428      |
| Shrink block 0 with max-pool, 16 to 24 to 32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | max__D0__Rprog       | 80.08    | 0.61 | 3/3   | target   | 438      |
| Shrink block 1 with bilinear, 16 to 24 to 32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | bilinear__D1__Rprog  | 80.23    | 0.38 | 3/3   | target   | 436      |
| Shrink block 1 with least-squares reconstruction, 16 to 24 to 32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | l2__D1__Rprog        | 80.11    | 0.52 | 3/3   | target   | 510      |
| Shrink block 1 with max-pool then anti-alias blur, 16 to 24 to 32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | maxblur__D1__Rprog   | 80.43    | 0.75 | 3/3   | target   | 519      |
| Shrink block 1 with max-pool then anti-alias blur, 16 to 24 to 32, longer at low res                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | maxblur__D1__Rlate   | 80.20    | 0.52 | 3/3   | target   | 446      |
| Shrink block 1 with max-pool then anti-alias blur, 24 to 16 to 32 (order control)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | maxblur__D1__Reverse | 80.46    | 0.44 | 3/3   | target   | 448      |
| Shrink block 1 with max-pool then anti-alias blur, 24 to 32 only                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | maxblur__D1__Rgentle | 80.32    | 0.22 | 3/3   | target   | 499      |
| Shrink block 1 with max-pool, 16 to 24 to 32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | max__D1__Rprog       | 80.60    | 0.20 | 3/3   | target   | 433      |
| Shrink block 1 with max-pool, 16 to 24 to 32, longer at low res                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | max__D1__Rlate       | 80.07    | 0.53 | 3/3   | target   | 436      |

continued on next page

| Configuration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Config id                             | Acc. (%) | SD   | Seeds | Endpoint          | Wall (s) |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|----------|------|-------|-------------------|----------|
| Shrink block 1 with max-pool, 24 to 16 to 32 (order control)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | max__D1__Rreverse                     | 80.21    | 0.30 | 3/3   | target            | 438      |
| Shrink block 1 with max-pool, 24 to 32 only                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | max__D1__Rgentle                      | 79.78    | 0.32 | 3/3   | target            | 430      |
| Shrink block 1 with max-pool, fixed 16x16, never restored ( <i>fixed-resolution control: trained and evaluated at 16x16 after block/site reduction; final metric is that reduced path, the 32x32 path is a diagnostic</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | max__D1__fixed16                      | 73.13    | 0.54 | 3/3   | current (control) | 419      |
| Shrink block 1 with max-pool, fixed 24x24, never restored ( <i>fixed-resolution control: trained and evaluated at 24x24; final metric is that reduced path</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | max__D1__fixed24                      | 76.15    | 0.49 | 3/3   | current (control) | 436      |
| Shrink block 1 with perceptual (SSIM-style), 16 to 24 to 32 [ <b>diverged: non-finite loss, seeds 0, 1, 2</b> ]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | perceptual__D1__Rprog                 | —        | —    | 0/3   | target            | —        |
| Shrink block 1 with smoothness-optimal reconstruction, 16 to 24 to 32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | hminus1__D1__Rprog                    | 79.85    | 0.76 | 3/3   | target            | 507      |
| Shrink block 1 with smoothness-optimal reconstruction, 16 to 24 to 32, longer at low res                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | hminus1__D1__Rlate                    | 79.55    | 0.66 | 3/3   | target            | 564      |
| Shrink block 1 with smoothness-optimal reconstruction, 24 to 16 to 32 (order control)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | hminus1__D1__Rreverse                 | 79.50    | 0.41 | 3/3   | target            | 527      |
| Shrink block 1 with smoothness-optimal reconstruction, 24 to 32 only                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | hminus1__D1__Rgentle                  | 78.90    | 0.11 | 3/3   | target            | 548      |
| Shrink block 1 with softmax-weighted pooling, 16 to 24 to 32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | softpool__D1__Rprog                   | 80.34    | 0.48 | 3/3   | target            | 482      |
| Shrink block 2 with max-pool, 16 to 24 to 32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | max__D2__Rprog                        | 80.24    | 0.56 | 3/3   | target            | 428      |
| Shrink the first conv layer with max-pool, 16 to 24 to 32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | max__stem__Rprog                      | 79.61    | 0.82 | 3/3   | target            | 409      |
| Shrink the input image with bilinear, 16 to 24 to 32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | bilinear__input__Rprog                | 79.56    | 0.68 | 3/3   | target            | 427      |
| Shrink the input image with least-squares reconstruction, 16 to 24 to 32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | l2__input__Rprog                      | 79.64    | 0.33 | 3/3   | target            | 537      |
| Shrink the input image with max-pool then anti-alias blur, 16 to 24 to 32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | maxblur__input__Rprog                 | 79.17    | 0.43 | 3/3   | target            | 505      |
| Shrink the input image with max-pool, 16 to 24 to 32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | max__input__Rprog                     | 79.23    | 0.48 | 3/3   | target            | 433      |
| Shrink the input image with perceptual (SSIM-style), 16 to 24 to 32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | perceptual__input__Rprog              | 79.98    | 0.40 | 3/3   | target            | 462      |
| Shrink the input image with smoothness-optimal reconstruction, 16 to 24 to 32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | hminus1__input__Rprog                 | 79.76    | 0.65 | 3/3   | target            | 560      |
| Shrink the input image with softmax-weighted pooling, 16 to 24 to 32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | softpool__input__Rprog                | 79.28    | 0.46 | 3/3   | target            | 456      |
| <b>21-configuration grid: resolution x blur x site</b> (campaign_grid21; owner Max; training benchmark)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                       |          |      |       |                   |          |
| Conditions: reference recipe; evaluation every 2 epochs. Evaluation split: test. Records: benchmark-organized@8d353f2; results committed in b9bb609. Knowledge-base record: EXP-011. Assets: r20bn-campaign-assets: shares initial weights, BatchNorm buffers, probe and per-epoch order, seed by seed, with every experiment on this set. Runs attempted per manifests: 63; with summary: 63; numerically valid: 63; diverged: 0; failure markers: 0; duplicate summaries collapsed: 0. Timing: wall_seconds from model construction to summary write, including evaluations and checkpoint writes; eval_seconds counts snapshot evaluation only; train_seconds = wall - eval (so it includes data indexing and checkpoint I/O). One T4, two cells in parallel on a two-GPU kernel. |                                       |          |      |       |                   |          |
| Baseline: no blur, full 32x32 throughout                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | R32__Gnone__input_bilinear__all19     | 75.62    | 0.49 | 3/3   | target            | 489      |
| Blend each layer between itself and its blur                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | R32__Gmix__input_bilinear__all19      | 78.18    | 0.36 | 3/3   | target            | 789      |
| Blur every conv layer, annealed to zero                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | R32__Gplateau__input_bilinear__all19  | 78.69    | 0.68 | 3/3   | target            | 696      |
| Blur every conv layer, annealed to zero, first 7 layers only                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | R32__Gplateau__input_bilinear__early7 | 75.24    | 0.74 | 3/3   | target            | 593      |
| Blur every conv layer, geometric decay                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | R32__Ggeo__input_bilinear__all19      | 78.28    | 0.50 | 3/3   | target            | 703      |
| Shrink the first conv layer (bilinear), 16 to 24 to 32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Rprog__Gnone__stem_bilinear__all19    | 78.91    | 0.37 | 3/3   | target            | 484      |
| Shrink the first conv layer (bilinear), 16 to 24 to 32, plus blur every conv layer, annealed to zero                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Rprog__Gplateau__stem_bilinear__all19 | 79.30    | 0.62 | 3/3   | target            | 750      |
| Shrink the first conv layer (max-pool), 16 to 24 to 32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Rprog__Gnone__stem_max__all19         | 79.97    | 0.58 | 3/3   | target            | 491      |
| Shrink the first conv layer (max-pool), 16 to 24 to 32, plus blur every conv layer, annealed to zero                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Rprog__Gplateau__stem_max__all19      | 79.81    | 0.79 | 3/3   | target            | 747      |

continued on next page

| Configuration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Config id                                 | Acc. (%) | SD   | Seeds | Endpoint          | Wall (s) |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|----------|------|-------|-------------------|----------|
| Shrink the input image (bilinear), 16 to 24 to 32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Rprog__Gnone__input_bilinear__all19       | 79.48    | 0.54 | 3/3   | target            | 492      |
| Shrink the input image (bilinear), 16 to 24 to 32, plus blend each layer between itself and its blur                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Rprog__Gmix__input_bilinear__all19        | 79.51    | 0.80 | 3/3   | target            | 783      |
| Shrink the input image (bilinear), 16 to 24 to 32, plus blur every conv layer, annealed to zero                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Rprog__Gplateau__input_bilinear__all19    | 80.71    | 0.65 | 3/3   | target            | 700      |
| Shrink the input image (bilinear), 16 to 24 to 32, plus blur every conv layer, annealed to zero, first 7 layers only                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Rprog__Gplateau__input_bilinear__early7   | 79.27    | 0.57 | 3/3   | target            | 596      |
| Shrink the input image (bilinear), 16 to 24 to 32, plus blur every conv layer, geometric decay                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Rprog__Ggeo__input_bilinear__all19        | 79.91    | 0.92 | 3/3   | target            | 709      |
| Shrink the input image (bilinear), 24 to 16 to 32 (order control)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Rreverse__Gnone__input_bilinear__all19    | 79.23    | 0.40 | 3/3   | target            | 467      |
| Shrink the input image (bilinear), 24 to 16 to 32 (order control), plus blur every conv layer, annealed to zero                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Rreverse__Gplateau__input_bilinear__all19 | 79.33    | 0.38 | 3/3   | target            | 757      |
| Shrink the input image (bilinear), 24 to 32 only                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Rgentle__Gnone__input_bilinear__all19     | 78.56    | 0.34 | 3/3   | target            | 486      |
| Shrink the input image (bilinear), 24 to 32 only, plus blur every conv layer, annealed to zero                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Rgentle__Gplateau__input_bilinear__all19  | 79.77    | 0.72 | 3/3   | target            | 694      |
| Shrink the input image (bilinear), 24 to 32 only, plus blur every conv layer, geometric decay                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Rgentle__Ggeo__input_bilinear__all19      | 79.68    | 0.57 | 3/3   | target            | 704      |
| Shrink the input image (max-pool), 16 to 24 to 32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Rprog__Gnone__input_max__all19            | 79.80    | 0.50 | 3/3   | target            | 489      |
| Shrink the input image (max-pool), 16 to 24 to 32, plus blur every conv layer, annealed to zero                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Rprog__Gplateau__input_max__all19         | 80.34    | 0.76 | 3/3   | target            | 738      |
| <b>Anti-aliasing ablation: placement, masks, constant sigma, BlurPool, internal reductions, per-layer profiles</b><br>(ablation_aa; owner Idriss; training benchmark)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                           |          |      |       |                   |          |
| Conditions: reference recipe; own pinned initial weights (r20bn-ablation-assets); evaluation every 2 epochs. Evaluation split: test. Records: continuation-gaussian-tv_exploration_1@591e125; results committed in efe24dc. Knowledge-base record: EXP-012 (branch continuation-gaussian-tv_exploration_1). Assets: r20bn-ablation-assets: same subset, probe and per-epoch order as r20bn-campaign-assets, different initial weights. Runs attempted per manifests: 58; with summary: 58; numerically valid: 58; diverged: 0; failure markers: 0; duplicate summaries collapsed: 0. Timing: wall_seconds from model construction to summary write, including evaluations and checkpoint writes; eval_seconds counts snapshot evaluation only; train_seconds = wall - eval (so it includes data indexing and checkpoint I/O). One T4, two cells in parallel on a two-GPU kernel. |                                           |          |      |       |                   |          |
| Baseline: no blur, full 32x32 throughout (ablation batch)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | C_plain                                   | 75.06    | n/a  | 1/1   | target            | 423      |
| Blur after every BatchNorm, annealed to zero                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | P_postbn                                  | 78.39    | n/a  | 1/1   | target            | 673      |
| Blur after every ReLU (10 places), annealed to zero                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | P_postblock                               | 80.08    | 0.21 | 3/3   | target            | 538      |
| Blur every conv layer, annealed to zero                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | C_plateau                                 | 78.18    | n/a  | 1/1   | target            | 648      |
| Blur only the 2 layers feeding a downsample                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | M_predown                                 | 77.55    | n/a  | 1/1   | target            | 436      |
| Blur rising with depth, after every ReLU (direction control)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Q_A3                                      | 77.01    | 0.38 | 3/3   | target            | 585      |
| Blur scaled by map size, after every ReLU                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Q_A1                                      | 78.48    | 0.33 | 3/3   | target            | 550      |
| Blur scaled by receptive field, after every ReLU                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Q_A4                                      | 79.59    | 0.10 | 3/3   | target            | 532      |
| Blur scaled by sqrt of map size, after every ReLU                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Q_A2                                      | 79.55    | 0.46 | 3/3   | target            | 579      |
| Blur the 17 layers away from a downsample                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | M_nodown                                  | 78.52    | n/a  | 1/1   | target            | 605      |
| Fixed anti-alias blur before each downsample (control) ( <i>fixed BlurPool architecture kept at inference: its target path is not the plain ResNet-20</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | B_blurpool                                | 78.51    | n/a  | 1/1   | target            | 460      |
| Fixed anti-alias blur before each downsample, plus annealed blur everywhere ( <i>fixed BlurPool architecture kept at inference: its target path is not the plain ResNet-20</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | B_blurpool_plateau                        | 79.19    | n/a  | 1/1   | target            | 698      |
| Fixed blur 0.30, never annealed (control) ( <i>fixed-filter control: constant sigma, never annealed; final metric is the filtered (current) path</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | K_const030                                | 74.97    | n/a  | 1/1   | current (control) | 755      |
| Fixed blur 0.50, never annealed (control) ( <i>fixed-filter control: constant sigma, never annealed; final metric is the filtered (current) path</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | K_const050                                | 78.69    | n/a  | 1/1   | current (control) | 738      |

continued on next page

| Configuration                                                                                                                                       | Config id  | Acc. (%) | SD   | Seeds | Endpoint          | Wall (s) |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------|------|-------|-------------------|----------|
| Fixed blur 0.80, never annealed (control)<br>(fixed-filter control: constant sigma, never annealed; final metric is the filtered (current) path)    | K_const080 | 75.02    | n/a  | 1/1   | current (control) | 736      |
| Fixed blur 1.00, never annealed (control)<br>(fixed-filter control: constant sigma, never annealed; final metric is the filtered (current) path)    | K_const100 | 71.15    | n/a  | 1/1   | current (control) | 749      |
| Fixed blur after every ReLU, never annealed (control) (fixed-filter control: constant post-ReLU sigma; final metric is the filtered (current) path) | R6         | 79.78    | n/a  | 1/1   | current (control) | 557      |
| Shrink after the first ReLU, plus blur after every ReLU                                                                                             | R5         | 79.65    | n/a  | 1/1   | target            | 536      |
| Shrink block 0 with max-pool, 16 to 24 to 32                                                                                                        | D0         | 80.44    | 0.43 | 3/3   | target            | 455      |
| Shrink block 0, plus blur after every ReLU                                                                                                          | D0G        | 80.53    | n/a  | 1/1   | target            | 545      |
| Shrink block 1 with max-pool, 16 to 24 to 32                                                                                                        | D1         | 80.59    | 0.41 | 3/3   | target            | 447      |
| Shrink block 1, plus blur after every ReLU                                                                                                          | D1G        | 80.08    | n/a  | 1/1   | target            | 519      |
| Shrink block 2 with max-pool, 16 to 24 to 32                                                                                                        | D2         | 80.09    | 0.17 | 3/3   | target            | 447      |
| Shrink block 2, plus blur after every ReLU                                                                                                          | D2G        | 80.99    | 0.41 | 3/3   | target            | 574      |
| Shrink block 2, plus blur rising with depth after every ReLU (direction control)                                                                    | P_A3       | 80.18    | 0.46 | 3/3   | target            | 583      |
| Shrink block 2, plus blur scaled by map size after every ReLU                                                                                       | P_A1       | 79.61    | 0.44 | 3/3   | target            | 633      |
| Shrink block 2, plus blur scaled by receptive field after every ReLU                                                                                | P_A4       | 81.14    | 0.22 | 3/3   | target            | 594      |
| Shrink block 2, plus blur scaled by sqrt of map size after every ReLU                                                                               | P_A2       | 80.28    | 0.24 | 3/3   | target            | 551      |
| Shrink the first conv layer, 16 to 24 to 32                                                                                                         | R4         | 79.95    | n/a  | 1/1   | target            | 437      |
| Shrink the input image, 16 to 24 to 32                                                                                                              | R1         | 78.92    | n/a  | 1/1   | target            | 443      |
| Shrink the input image, 16 to 24 to 32, plus blur after every ReLU                                                                                  | R3         | 80.14    | n/a  | 1/1   | target            | 542      |
| Shrink the input image, 16 to 24 to 32, plus blur every conv layer                                                                                  | R2         | 79.15    | n/a  | 1/1   | target            | 604      |

**Per-layer sigma profiles (rho), adaptive controller, sigma0 sweep (per\_layer\_sigma; owner Aubin; training benchmark)**

Conditions: own self-paired assets per seed (different digest scheme); per\_layer\_cpu is a 6,000/2,000-image CPU pilot of 12 epochs. Evaluation split: test. Records: benchmark-organized@8d353f2; adaptative-schedule@ea186fe; results committed in 325ea45, ea186fe. Knowledge-base record: EXP-012 (branch continuation-gaussian-tv); EXP-013 (branch adaptative-schedule). Assets: None: no pinned assets recorded; per-layer-self-paired: self-paired within the study; its digests use another scheme, so pairing with the other sets is not established. Runs attempted per manifests: not recorded; with summary: 49; numerically valid: 49; diverged: 0; failure markers: 0; duplicate summaries collapsed: 42. Timing: wall\_seconds from model construction to summary write, including evaluations and checkpoint writes; eval\_seconds counts snapshot evaluation only; train\_seconds = wall - eval (so it includes data indexing and checkpoint I/O). One T4, two cells in parallel on a two-GPU kernel.

|                                                                                                |                |       |      |     |        |   |
|------------------------------------------------------------------------------------------------|----------------|-------|------|-----|--------|---|
| Baseline: no blur, full 32x32 throughout (per-layer batch) [5000 train / 2000 eval, 4 epochs]  | plain          | 28.40 | n/a  | 1/1 | target | — |
| Baseline: no blur, full 32x32 throughout (per-layer batch) [6000 train / 2000 eval, 12 epochs] | plain          | 47.65 | n/a  | 1/1 | target | — |
| Baseline: no blur, full 32x32 throughout (per-layer batch) [launch per-layer-adaptive-fix]     | plain          | 73.91 | 0.62 | 3/3 | target | — |
| Baseline: no blur, full 32x32 throughout (per-layer batch) [launch per-layer-sigma]            | plain          | 74.42 | 0.29 | 3/3 | target | — |
| Baseline: no blur, full 32x32 throughout (per-layer batch), sigma0 = 1                         | plain@sigma0=1 | 74.13 | 0.73 | 3/3 | target | — |
| Per-layer blur strength chosen by the network [5000 train / 2000 eval, 4 epochs]               | adaptive       | 31.65 | n/a  | 1/1 | target | — |
| Per-layer blur strength chosen by the network [launch per-layer-adaptive-fix]                  | adaptive       | 77.35 | 0.49 | 3/3 | target | — |
| Per-layer blur strength chosen by the network [launch per-layer-sigma]                         | adaptive       | 77.50 | 0.61 | 3/3 | target | — |
| Per-layer blur strength, exponent 0.5                                                          | rho0.5         | 74.65 | 0.67 | 3/3 | target | — |
| Per-layer blur strength, exponent 0.5 [6000 train / 2000 eval, 12 epochs]                      | rho0.5         | 45.80 | n/a  | 1/1 | target | — |
| Per-layer blur strength, exponent 1 [6000 train / 2000 eval, 12 epochs]                        | rho1           | 49.00 | n/a  | 1/1 | target | — |
| Per-layer blur strength, exponent 1 [launch per-layer-adaptive-fix]                            | rho1           | 77.66 | 0.68 | 3/3 | target | — |

continued on next page

| Configuration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Config id           | Acc. (%) | SD   | Seeds | Endpoint | Wall (s) |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|----------|------|-------|----------|----------|
| Per-layer blur strength, exponent 1 [launch per-layer-sigma]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | rho1                | 77.65    | 0.74 | 3/3   | target   | —        |
| Per-layer blur strength, exponent 1, sigma0 = 0.25                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | s0_0.25@sigma0=0.25 | 73.81    | 0.96 | 3/3   | target   | —        |
| Per-layer blur strength, exponent 1, sigma0 = 0.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | s0_0.5@sigma0=0.5   | 76.41    | 0.86 | 3/3   | target   | —        |
| Per-layer blur strength, exponent 1, sigma0 = 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | rho1@sigma0=1       | 77.84    | 0.63 | 3/3   | target   | —        |
| Per-layer blur strength, exponent 1, sigma0 = 1.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | s0_1.5@sigma0=1.5   | 74.55    | 0.38 | 3/3   | target   | —        |
| Per-layer blur strength, exponent 1, sigma0 = 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | s0_2@sigma0=2       | 70.59    | 0.71 | 3/3   | target   | —        |
| Per-layer blur strength, exponent 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | rho2                | 76.65    | 0.21 | 3/3   | target   | —        |
| Per-layer blur strength, exponent 2 [5000 train / 2000 eval, 4 epochs]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | rho2                | 31.80    | n/a  | 1/1   | target   | —        |
| Per-layer blur strength, exponent 2 [6000 train / 2000 eval, 12 epochs]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | rho2                | 50.10    | n/a  | 1/1   | target   | —        |
| <b>Data-triggered schedules, gradient-norm calibration, dwell allocation, long-horizon runs (adaptive_continuation; owner Aubin; training benchmark)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                     |          |      |       |          |          |
| Conditions: reference recipe, mostly seed 0 only; long-conv runs use 120 epochs and are not comparable on accuracy. Evaluation split: test. Records: benchmark-organized@8d353f2; results committed in 325ea45, 9c4492e. Knowledge-base record: docs/experiments_2026-09-10.md, docs/adaptive_continuation_maths.md. Assets: <b>None</b> : no pinned assets recorded; <b>r20bn-campaign-assets</b> : shares initial weights, BatchNorm buffers, probe and per-epoch order, seed by seed, with every experiment on this set. Runs attempted per manifests: 16; with summary: 15; numerically valid: 15; diverged: 0; failure markers: 1; duplicate summaries collapsed: 0. Timing: wall_seconds from model construction to summary write, including evaluations and checkpoint writes; eval_seconds counts snapshot evaluation only; train_seconds = wall - eval (so it includes data indexing and checkpoint I/O). One T4, two cells in parallel on a two-GPU kernel. |                     |          |      |       |          |          |
| adaptgap hi [50000 train / 10000 eval, 120 epochs] ( <i>budget 120 epochs: not comparable on accuracy with 30-epoch arms</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | adaptgap_hi         | 86.73    | n/a  | 1/1   | target   | 2639     |
| adaptgap lo [50000 train / 10000 eval, 120 epochs] ( <i>budget 120 epochs: not comparable on accuracy with 30-epoch arms</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | adaptgap_lo         | 78.40    | n/a  | 1/1   | target   | 2598     |
| Blur every conv layer, annealed (instrumented rerun) [launch gradnorm-cal]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Gplateau_cal        | 77.97    | n/a  | 1/1   | target   | 637      |
| Blur every conv layer, annealed (instrumented rerun) [launch gradnorm-cal2]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Gplateau_cal        | 77.94    | n/a  | 1/1   | target   | 719      |
| Blur every conv layer, annealed (instrumented rerun) [launch signals-cal3]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Gplateau_cal        | 78.33    | n/a  | 1/1   | target   | 721      |
| Blur schedule chosen by a transfer-gap trigger (mis-set) [launch adaptive-gap]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | ADAPTgap            | 75.39    | n/a  | 1/1   | target   | 619      |
| Blur schedule chosen by a transfer-gap trigger [launch adaptive-gap2]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | ADAPTgap            | 77.13    | n/a  | 1/1   | target   | 622      |
| Blur schedule chosen by the network (first trigger)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | ADAPT               | 76.34    | n/a  | 1/1   | target   | 610      |
| Blur schedule, more time at high blur                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | alloc_front         | 76.57    | n/a  | 1/1   | target   | 663      |
| Blur schedule, more time at low blur                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | alloc_back          | 78.74    | n/a  | 1/1   | target   | 654      |
| gplateau hi [50000 train / 10000 eval, 120 epochs] ( <i>budget 120 epochs: not comparable on accuracy with 30-epoch arms</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | gplateau_hi         | 87.16    | n/a  | 1/1   | target   | 2594     |
| gplateau lo [50000 train / 10000 eval, 120 epochs] ( <i>budget 120 epochs: not comparable on accuracy with 30-epoch arms</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | gplateau_lo         | 80.68    | n/a  | 1/1   | target   | 2765     |
| plain hi [50000 train / 10000 eval, 120 epochs] ( <i>budget 120 epochs: not comparable on accuracy with 30-epoch arms</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | plain_hi            | 86.04    | n/a  | 1/1   | target   | 1752     |
| plain lo [50000 train / 10000 eval, 120 epochs] ( <i>budget 120 epochs: not comparable on accuracy with 30-epoch arms</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | plain_lo            | 77.10    | n/a  | 1/1   | target   | 1774     |
| Resolution schedule chosen by a transfer-gap trigger                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | ADAPTRES            | 77.29    | n/a  | 1/1   | target   | 410      |
| <b>Full-data plain / Gaussian plateau / geometric (fulldata_gaussian; owner Max; training benchmark)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                     |          |      |       |          |          |
| Conditions: reference recipe; evaluation every 2 epochs; this run produced the pinned campaign asset set. Evaluation split: test. Records: benchmark-organized@8d353f2; results committed in d643f75. Knowledge-base record: EXP-008. Assets: <b>r20bn-campaign-assets</b> : shares initial weights, BatchNorm buffers, probe and per-epoch order, seed by seed, with every experiment on this set. Runs attempted per manifests: not recorded; with summary: 9; numerically valid: 9; diverged: 0; failure markers: 0; duplicate summaries collapsed: 0. Timing: wall_seconds of the run as recorded by continuation_driver; includes periodic evaluation.                                                                                                                                                                                                                                                                                                           |                     |          |      |       |          |          |
| geometric_seed0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | geometric           | 78.46    | 0.72 | 3/3   | target   | 684      |
| plain_seed0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | plain               | 75.40    | 0.40 | 3/3   | target   | 458      |
| plateau_seed0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | plateau             | 78.41    | 0.32 | 3/3   | target   | 693      |

continued on next page

| Configuration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Config id             | Acc. (%) | SD   | Seeds | Endpoint | Wall (s) |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|----------|------|-------|----------|----------|
| <b>Progressive input resolution, one seed (progressive_resolution_pilot; owner Max; training pilot)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                       |          |      |       |          |          |
| Conditions: reference recipe, seed 0; pairs with fulldata_gaussian seed 0. Evaluation split: test. Records: benchmark-organized@8d353f2; results committed in 26cfb82. Knowledge-base record: EXP-010. Assets: r20bn-campaign-assets: shares initial weights, BatchNorm buffers, probe and per-epoch order, seed by seed, with every experiment on this set. Runs attempted per manifests: not recorded; with summary: 2; numerically valid: 2; diverged: 0; failure markers: 0; duplicate summaries collapsed: 0. Timing: wall_seconds of the run as recorded by continuation_driver; includes periodic evaluation. |                       |          |      |       |          |          |
| progres_gauss_seed0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | progres_gauss         | 79.93    | n/a  | 1/1   | target   | 674      |
| progres_seed0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | progres               | 79.57    | n/a  | 1/1   | target   | 452      |
| <b>ResNet-20 BN pilot, plain vs internal Gaussian (resnet20bn_gaussian_pilot; owner Max; training pilot)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                       |          |      |       |          |          |
| Conditions: 10,000-image subset, 2,400 updates, val split; seed 0. Evaluation split: val. Records: benchmark-organized@8d353f2; results committed in d643f75. Knowledge-base record: EXP-007. Assets: None: no pinned assets recorded. Runs attempted per manifests: not recorded; with summary: 2; numerically valid: 2; diverged: 0; failure markers: 0; duplicate summaries collapsed: 0. Timing: wall_seconds of the run as recorded by continuation_driver; includes periodic evaluation.                                                                                                                       |                       |          |      |       |          |          |
| gaussian_r20bn_seed0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | gaussian_r20bn        | 59.62    | n/a  | 1/1   | target   | 174      |
| plain_r20bn_seed0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | plain_r20bn           | 55.36    | n/a  | 1/1   | target   | 119      |
| <b>ResNet-18 BN, plain vs internal Gaussian (resnet18bn_gaussian_pilot; owner Max; training pilot)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                       |          |      |       |          |          |
| Conditions: 10,000-image subset, val split; ResNet-18 BatchNorm; seed 0. Evaluation split: val. Records: benchmark-organized@8d353f2; results committed in d643f75. Knowledge-base record: EXP-006. Assets: None: no pinned assets recorded. Runs attempted per manifests: not recorded; with summary: 3; numerically valid: 3; diverged: 0; failure markers: 0; duplicate summaries collapsed: 0. Timing: wall_seconds of the run as recorded by continuation_driver; includes periodic evaluation.                                                                                                                 |                       |          |      |       |          |          |
| gaussian_r18_seed0 [launch resnet18-gaussian]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | gaussian_r18          | 62.16    | n/a  | 1/1   | target   | 197      |
| gaussian_r18_seed0 [launch resnet18-gaussian-complete]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | gaussian_r18          | 62.16    | n/a  | 1/1   | target   | 197      |
| plain_r18_seed0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | plain_r18             | 54.26    | n/a  | 1/1   | target   | 151      |
| <b>db2 wavelet shrinkage pilot (thread closed) (db2_pilot; owner Max; training pilot)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                       |          |      |       |          |          |
| Conditions: 10,000-image subset, 2,400 updates, val split; seed 0. Evaluation split: val. Records: benchmark-organized@8d353f2; results committed in 26cfb82. Knowledge-base record: EXP-009. Assets: None: no pinned assets recorded. Runs attempted per manifests: not recorded; with summary: 2; numerically valid: 2; diverged: 0; failure markers: 0; duplicate summaries collapsed: 0. Timing: wall_seconds of the run as recorded by continuation_driver; includes periodic evaluation.                                                                                                                       |                       |          |      |       |          |          |
| db2_r20bn_seed0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | db2_r20bn             | 56.34    | n/a  | 1/1   | target   | 3856     |
| plain_r20bn_seed0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | plain_r20bn           | 55.40    | n/a  | 1/1   | target   | 89       |
| <b>GroupNorm LR diagnosis and 10,000-image plain vs Gaussian (gn_lr_audit; owner Max; training pilot)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                       |          |      |       |          |          |
| Conditions: 10,000 train images, 5,000 val; ResNet-20 GroupNorm; 1,200 updates; val accuracy, not test. Evaluation split: val. Records: benchmark-organized@8d353f2; results committed in d643f75. Knowledge-base record: EXP-005. Assets: None: no pinned assets recorded. Runs attempted per manifests: not recorded; with summary: 10; numerically valid: 10; diverged: 0; failure markers: 0; duplicate summaries collapsed: 0. Timing: wall_seconds of the run as recorded by continuation_driver; includes periodic evaluation.                                                                                |                       |          |      |       |          |          |
| gaussian_seed0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | gaussian              | 46.91    | 0.60 | 3/3   | target   | 61       |
| gaussian_seed0_lr0p02                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | gaussian_seed0_lr0p02 | 47.94    | n/a  | 1/1   | target   | 66       |
| plain_lr0p005                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | plain_lr0p005         | 41.78    | n/a  | 1/1   | target   | 25       |
| plain_lr0p02                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | plain_lr0p02          | 34.74    | n/a  | 1/1   | target   | 26       |
| plain_seed0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | plain                 | 50.36    | 0.52 | 3/3   | target   | 47       |
| plain_seed0_lr0p02                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | plain_seed0_lr0p02    | 50.72    | n/a  | 1/1   | target   | 49       |
| <b>First internal-filter pilots (prelim checks, one collapsed run) (pilot_internal_gaussian; owner Max; training pilot)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                       |          |      |       |          |          |
| Conditions: 10,000-image subsets, ResNet-20 GroupNorm; the first pilot collapsed to uniform prediction (LR too high). Evaluation split: val. Records: benchmark-organized@8d353f2; results committed in d643f75. Knowledge-base record: EXP-004. Assets: None: no pinned assets recorded. Runs attempted per manifests: not recorded; with summary: 5; numerically valid: 5; diverged: 0; failure markers: 1; duplicate summaries collapsed: 0. Timing: wall_seconds of the run as recorded by continuation_driver; includes periodic evaluation.                                                                    |                       |          |      |       |          |          |
| db2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | db2                   | 23.34    | n/a  | 1/1   | target   | 648      |
| gaussian [launch pilot-continuation]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | gaussian              | 20.26    | n/a  | 1/1   | target   | 33       |
| gaussian [launch pilot-manual]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | gaussian              | 20.62    | n/a  | 1/1   | target   | 31       |
| plain [launch pilot-continuation]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | plain                 | 14.10    | n/a  | 1/1   | target   | 24       |
| plain [launch pilot-manual]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | plain                 | 14.12    | n/a  | 1/1   | target   | 23       |
| <b>Fixed Gaussian input blur, GroupNorm (exp0_input_gaussian; owner Max; training pilot)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                       |          |      |       |          |          |
| Conditions: CIFAR-10 45,000 train / 5,000 stratified val; ResNet-20 GroupNorm; 14,040 updates; fixed sigma per run. Evaluation split: val. Records: benchmark-organized@8d353f2; results committed in 86c00bc. Knowledge-base record: EXP-000. Assets: None: no pinned assets recorded. Runs attempted per manifests: not recorded; with summary: 15; numerically valid: 15; diverged: 0; failure markers: 0; duplicate summaries collapsed: 0. Timing: not recorded                                                                                                                                                 |                       |          |      |       |          |          |
| level_0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | level_0               | 84.11    | 0.60 | 3/3   | target   | —        |
| level_0p5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | level_0p5             | 82.55    | 0.06 | 3/3   | target   | —        |
| level_1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | level_1               | 71.33    | 2.38 | 3/3   | target   | —        |
| level_2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | level_2               | 48.32    | 2.96 | 3/3   | target   | —        |
| level_3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | level_3               | 38.85    | 1.86 | 3/3   | target   | —        |
| <b>Input-blur warm starts, GroupNorm (exp1_input_warmstart; owner Max; training pilot)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                       |          |      |       |          |          |
| Conditions: as exp0; warm-start branching from blurred prefixes. Evaluation split: val. Records: benchmark-organized@8d353f2; results committed in 86c00bc. Knowledge-base record: EXP-001. Assets: None: no pinned assets recorded. Runs attempted per manifests: not recorded; with summary: 9; numerically valid: 9; diverged: 0; failure markers: 0; duplicate summaries collapsed: 0. Timing: not recorded                                                                                                                                                                                                      |                       |          |      |       |          |          |
| arm_P                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | arm_P                 | 83.19    | 0.51 | 3/3   | target   | —        |
| arm_W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | arm_W                 | 82.93    | 0.11 | 3/3   | target   | —        |
| prefix_sigma_warm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | prefix_sigma_warm     | 55.57    | 2.10 | 3/3   | target   | —        |

continued on next page

| Configuration                                                                                                                                                                                                                                                                                                                             | Config id | Acc. (%) | SD | Seeds | Endpoint | Wall (s) |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------|----|-------|----------|----------|
| <b>Records without training</b>                                                                                                                                                                                                                                                                                                           |           |          |    |       |          |          |
| <b>TV-L2 / <math>\text{TV-}H^{-1}</math> budget projections (image previews)</b> (tv_budget_previews; owner Max): ten CIFAR-10 images, Chambolle-Pock solves; no classifier was trained. Unrelated to the resbench 'l2' / 'hminus1' reduction operators, which are constrained quadratic down-sampling layers trained inside the network. |           |          |    |       |          |          |
| <b>Undecimated wavelet shrinkage previews and cost</b> (wavelet_previews; owner Max): image previews and micro-benchmarks; no classifier trained.                                                                                                                                                                                         |           |          |    |       |          |          |
| <b>Loss-landscape slices around a trained Gaussian checkpoint</b> (loss_landscape_blur; owner Aubin): 2-D filter-normalised slices, filtered vs bypassed objective, optional BN recalibration; no training.                                                                                                                               |           |          |    |       |          |          |
| <b>Kaggle accelerator and mount probes</b> (infrastructure_probes; owner Max): environment probes only.                                                                                                                                                                                                                                   |           |          |    |       |          |          |